



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions. (10x3=30)

1. Does Fermi energy depend on the size (or volume) of the conductor? Give formula.
2. What is physical meaning of Fermi temperature?
3. What is Fermi wave vector? Derive its expression.
4. What is meant by energy gap and why such gaps appear due to Bragg's reflection at zone boundary?
5. Define Bloch function. What does it represent physically?
6. State some major failures of free electron model.
7. Distinguish between intrinsic and extrinsic semiconductors.
8. How can the effective masses of electrons and holes be determined by the technique of cyclotron frequency?
9. What are acceptor and donor states?
10. Why the product np is constant at a given temperature in case of semiconductors?

Answer the following questions.

Q. 2. (a): Derive expressions for dispersion relations and Fermi energy for 3-dimensional free electron Fermi gas under periodic boundary conditions.

(5)

(b): Prove that kinetic energy of three dimensional gas of N free electrons at $0K$ is

$$U_0 = \frac{3}{5} N \epsilon_F. \quad (5)$$

Q. 3. (a): Give postulates of Free Electron Model. In what respect this model failed? (5)

(b): How can you establish the origin of energy gap using the idea of the standing waves? (5)

Q. 4. (a): Using the semi-classical model, show that the electron moves on a surface of constant energy in k -space. (5)

(b): Define and derive the expression for the effective mass of an electron. (5)

→ BS-4 years Program / Six semester
Spring- 2023
Solid state physics-II (3402)

Q NO# 01:- Answer the following
short Question:

~ (a) i ~

→ Fermi energy:

Fermi energy is,

$$E_F = \frac{h^2}{2m} \left(\frac{3\pi^2 N}{V} \right)^{2/3}$$

So,

the fermi energy depends only
on the particle concentration.

$$E_F = \frac{h^2}{2m \times 4\pi^2} \left(\frac{3\pi^2 N}{V} \right)^{2/3}$$

$$= \frac{h^2}{2m} \left(\frac{3\pi^2 N}{2^3 \pi^3 V} \right)^{2/3}$$

$$E_F = \frac{h^2}{2m} \left(\frac{3N}{8\pi V} \right)^{2/3}$$

$$\therefore n = \frac{N}{V}$$

$$E_F = \frac{h^2}{2m} \left(\frac{3n}{8\pi} \right)^{2/3}$$

The fermi energy is indepen-
dent of size and volume of
conductor as it only depends

on number of electrons per unit volume or electron concentration.

ii

→ Physical meaning of Fermi temperature:

The Fermi temperature is not the temperature of electron gas. It is a measure of where the Fermi energy is at (typically on order of 10000K). So for most metals, at room temperature not many electrons are excited above the Fermi energy.

iii

→ Fermi wave vector:

The total number of energy orbitals in a sphere of radius k_F which should be equal to N . The volume element in the k -space (volume per single set of

k_x, k_y and k_z) is equal to $(2\pi/L)^3$.

Thus, in the sphere of volume
 $V = \frac{4}{3} \pi k_F^3$, the total number
of states is,

$$N = 2 \times \frac{\frac{4}{3} \pi k_F^3}{(2\pi/L)^3}$$

$$= 2 \times \frac{4}{3} \pi k_F^3 \cdot \left(\frac{L}{2\pi}\right)^3$$

$$= \frac{8}{3} \pi k_F^3 \cdot \left(\frac{L}{2\pi}\right)^3$$

$$N = \frac{8}{3} \pi k_F^3 \cdot \frac{L^3}{8\pi^3} \because L^3 = V$$

$$= \frac{L^3 k_F^3}{3\pi^2}$$

$$N = \frac{V k_F^3}{3\pi^2}$$

The number 2 is multiplied

since each orbital has two
allowed values of spin quantum
number m_s for each allowed
value of k .

$$k_F^3 = \frac{3\pi^2 N}{V}$$

$$k_F = \left(\frac{3\pi^2 N}{V}\right)^{1/3}$$

div

→ When electrons are exposed to a periodic potential gaps arise in their dispersion relation at the Brillouin zone boundary. The dispersion is quadratic approaching a zone boundary. Thus, the electronic spectrum breaks into bands, with forbidden **energy gaps** between the bands. In the nearly free electron model, the gaps are proportional to the periodic potential.

Bragg reflection is a characteristic feature of wave propagation in crystals. Bragg reflection of electron waves in crystals is the cause of energy gaps.

div

→ **Bloch Function:**

PUACP

According to Bloch, solution of the Schrodinger

equation for a periodic potential must be of a special form:

$$\psi_k(x) = e^{ikx} u_k(x)$$

The periodic function $u_k(x)$ is usually known as a Bloch function and $\psi_k(x)$ is sometimes known as a modified plane wave.

Bloch's theorem guarantees that all eigen states are some periodic function times a plane wave.

In repeated zone scheme the wave vector (the crystal momentum) can always be taken in the first Brillouin zone.

In fact $\hbar\vec{k}$ cannot be the momentum of a Bloch wave because the states do not change if we add or subtract a reciprocal lattice vector. However, $\hbar\vec{k}$ is still a useful quantity because conservation rules for $\vec{k}$ applies for scattering process in solids. Instead of simply momentum, $\hbar\vec{k}$ it is called the crystal momentum.

~(vi)~

→ Failures of free electron model:

The classical theory could not explain the following properties of metals.

- (i) Specific heat
- (ii) Para magnetism
- (iii) Long mean path at low pressure
- (iv) Linear variation of resistivity with temperature.
- (v) The distinction between metals, semimetals, semiconductors and insulator
- (vi) The occurrence of positive values of the Hall coefficient
- (vii) The relation of conduction electrons in the metal to the valence electrons of free atom.
- (viii) Many transport properties, particularly magneto transport.

PUACP

Q. vii Distinguish between intrinsic and extrinsic semiconductors.

→ Intrinsic Semiconductor

- Semiconductors in the pure form having no impurity are called intrinsic semiconductors.

Example:

- Silicon and germanium.
- Si and Ge in pure form have very low conductivity. i.e. adding small amount of impurity to Si or Ge conductivity increase & there is large no. of free charge carriers.

→ Extrinsic Semiconductor

- Semiconductors in which some impurity has been added are called extrinsic semiconductors.

Example:

- donor atoms are P, As and Sb.
- Acceptor atoms are B, Al and Ga.
- At absolute zero of temperature electrons are bound in covalent bonds in semiconductors. Hence conductivity of semiconductors increase as temperature increase.

Q. VIII

Answer: In many semiconductors it has been possible to determine by cyclotron resonance the effective masses of carriers in the conduction and valence bands near the band edges.

When the semiconductor is placed in a static magnetic field B , the electron move on circular orbits around the axis of the field.

The angular frequency of this motion calculated by equating the Lorentz force with the centripetal force.

$$\frac{m^* v^2}{r} = \left| -\frac{e}{c} \vec{v} \times \vec{B} \right|$$

$$\because \sin 90^\circ = 1$$

$$= \frac{e}{c} v B \sin 90^\circ$$

$$\frac{m^* v^2}{r} = \frac{e}{c} v B \Rightarrow \frac{m^* v}{r} = \frac{e B}{c}$$

$$\because v = r \omega_c$$

$$\frac{v}{r} = \frac{e B}{m^* c} \Rightarrow \frac{r \omega_c}{r} = \frac{e B}{m^* c}$$

$$\omega_c = \frac{e B}{m^* c}$$

Here m^* is appropriate cyclotron effective mass.

Section IX

Accepter & Donor:-

Donors:- Consider the example of arsenic atom. Arsenic have five valence electrons but silicon have only four valence electrons. Thus four electrons on the arsenic form tetrahedral covalent bonds similar to silicon, and the fifth electron is available for conduction. The arsenic atom is called a donor because when ionized it donates an electron to the conduction band.

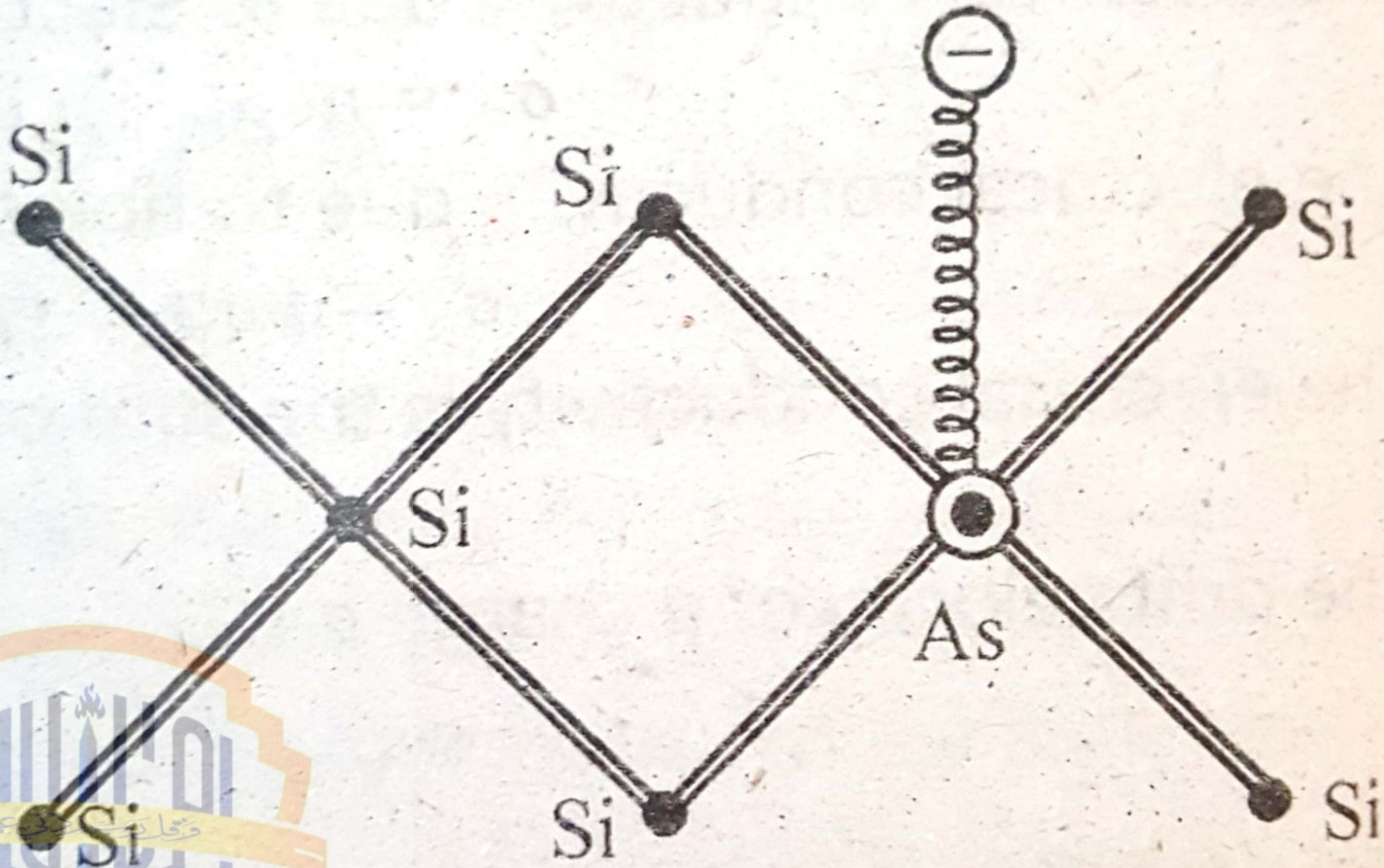
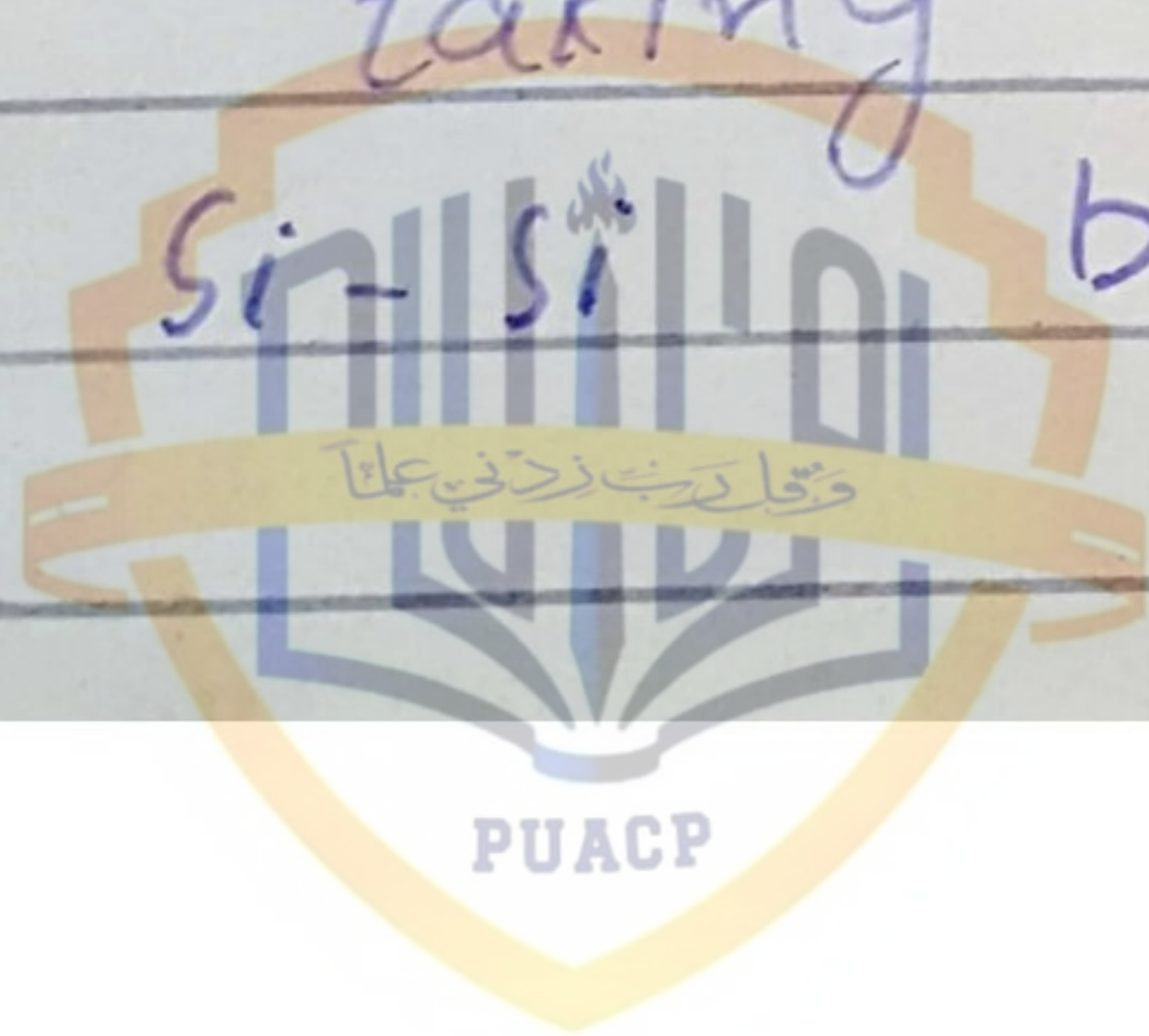


Fig 9.14: Charges associated with an arsenic impurity atom in silicon

Acceptors:- Consider the example of boron. Boron has only three valance electrons, it can complete its tetrahedral bonds only by taking an electron from a Si-Si bond, leaving behind a



hole in the silicon valence band.

The positive hole is then available for conduction. The boron atom is called an acceptor because when ionized it accepts an electron from the valence band.

At ^{OK} the hole is bound.

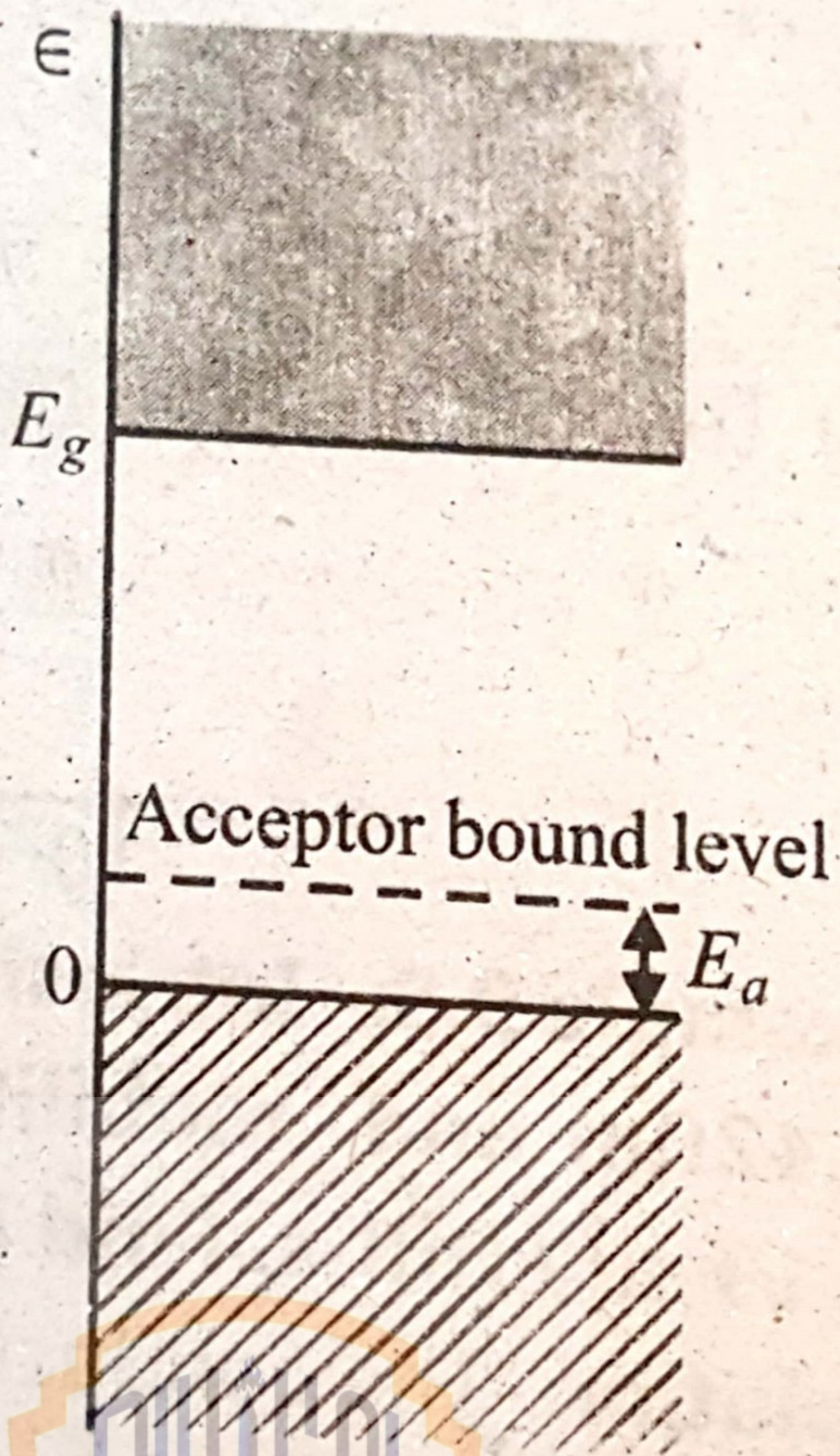


Fig 9.15: Acceptor level

PUACP

Long Question

Question No # 02:-

- (a) page no # 193 (ilmi)
- (b) Page no # 196 (ilmi)

Question No # 03:-

- (a) page no # 220 (ilmi)
- (b) page no # 223 (ilmi)

Question No # 04:-

- (a) page no # 248 (ilmi)
- (b) page no # 249 (ilmi)

Good Luck 😊

